Mathematical and Computational Methods

Exercise Sheet 1: Foundations: numbers, sets and functions

This sheet accompanies *Unit 1 (Foundations: numbers, sets and functions)*. Every problem uses only the methods of Unit 1—set and interval notation, the summation and product symbols, and the elementary functions; no calculus, vectors or complex numbers are needed.

How to use this sheet. Problems are graded by difficulty—⋆ drill (routine fluency), ⋆⋆ core (standard, tutorial level) and ⋆⋆⋆ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ▶) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showsolutions{}\input{MCM_Exercises_Unit1}"`) and recompile.

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1 Warm-up drills

Short routine problems—at least one per skill—to build fluency.

Exercise 1.1 *(⋆ drill — intervals & sets)*.

(a) Write $\{ x \in \mathbb{R} \ : \ -2 \le x < 5 \}$ in interval notation.

(b) Write $(-\infty, 3]$ in set-builder notation.

(c) Express $\{ x \in \mathbb{R} \ : \ x^2 \le 9 \}$ as a single interval.

(d) Compute $(-1, 4] \cap [2, 7)$.

(e) Compute $[0, 5] \setminus (2, 3)$.

Exercise 1.2 *(⋆ drill — Σ and Π notation)*. Evaluate

$$\text{(a)} \sum_{k=1}^{4} (3k-2), \qquad \text{(b)} \sum_{k=1}^{5} k^2, \qquad \text{(c)} \prod_{k=1}^{4} k, \qquad \text{(d)} \prod_{k=2}^{4} \left(1 - \tfrac{1}{k}\right).$$

Exercise 1.3 *(⋆ drill — inequalities)*. Solve for real x and give the answer as an interval (or union of intervals).

$$\text{(a) } 3 - 2x \le 7, \qquad \text{(b) } x^2 - x - 6 > 0, \qquad \text{(c) } |2x - 1| \le 3, \qquad \text{(d) } |x + 2| > 1.$$

Exercise 1.4 *(⋆ drill — maximal domain)*. State the maximal (natural) domain of each function.

$$\text{(a) } f(x) = \frac{1}{x-2}, \quad \text{(b) } g(x) = \sqrt{x-4}, \quad \text{(c) } h(x) = \ln(x+1), \quad \text{(d) } k(x) = \frac{1}{\sqrt{3-x}}.$$

Exercise 1.5 *(⋆ drill — composition)*. Let $f(x) = 2x + 1$ and $g(x) = x^2$. Find $(f \circ g)(x)$, $(g \circ f)(x)$ and $(f \circ f)(x)$, and evaluate $(f \circ g)(3)$.

Exercise 1.6 *(⋆ drill — injective / surjective / bijective)*. For each map $f\colon \mathbb{R} \to \mathbb{R}$, state whether it is injective, surjective, both (bijective) or neither.

$$\text{(a) } f(x) = 3x - 1, \quad \text{(b) } f(x) = x^2, \quad \text{(c) } f(x) = x^3, \quad \text{(d) } f(x) = e^x.$$

Exercise 1.7 *(⋆ drill — inverse)*. Find the inverse of each function (with its domain).

$$\text{(a) } f(x) = 5x + 2 \text{ on } \mathbb{R}, \qquad \text{(b) } f(x) = \sqrt{x} + 1 \text{ on } [0, \infty).$$

Exercise 1.8 *(⋆ drill — even / odd / periodic)*. Classify each as even, odd, or neither; for (d) also state the period.

$$\text{(a) } x^4 - 3x^2, \quad \text{(b) } x^3 + x, \quad \text{(c) } x^2 + x, \quad \text{(d) } \cos(2x).$$

Exercise 1.9 *(⋆ drill — graph transformations)*. Describe in words how the graph of $y = f(x)$ moves to give (a) $y = f(x) + 3$, (b) $y = f(x - 2)$, (c) $y = -f(x)$, (d) $y = f(2x)$. Then, starting from $y = x^2$, identify the transformations giving $y = (x + 1)^2 - 4$ and state its vertex.

Exercise 1.10 *(⋆ drill — hyperbolic functions)*.

(a) Evaluate $\sinh 0$, $\cosh 0$ and $\tanh 0$ from the definitions.

(b) Is sinh even or odd? Justify from $\sinh x = \frac{1}{2}(e^x - e^{-x})$.

(c) Without a calculator, what is the smallest possible value of $\cosh x$, and where is it attained?

Exercise 1.11 *(⋆ drill — the graphs of* tan *and* cot*)*. On separate axes, sketch $y = \tan x$ and $y = \cot x$ over $[-\pi, \pi]$. Mark every vertical asymptote and every x-intercept in that range, and state the period of each.

2 Tutorial problems

Standard-difficulty problems, anchored on the unit's worked examples, with several flavoured for the different cohorts (mathematics, data science, physics).

Problems marked ▶ are the core set for the 2-hour tutorial; the rest are for completion at home.

Exercise 2.1 *(▶ ⋆⋆ core — maximal domain and range)*.

(a) Find the maximal domain of $f(x) = \dfrac{\sqrt{x+3}}{x-1}$.

(b) Find both the maximal domain *and* the range of $g(x) = 2 - \sqrt{4 - x}$.

Exercise 2.2 *(▶ ⋆⋆ core — inequalities, two ways)*. Solve

$$\text{(a) } \frac{2x+1}{x-3} \geq 0, \qquad \text{(b) } |x - 1| < |x + 3|.$$

Exercise 2.3 *(⋆⋆ core — Σ manipulation (maths))*.

(a) Using $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, show that $\displaystyle\sum_{k=1}^{n}(3k + 2) = \frac{3n^2 + 7n}{2}$, and evaluate it at $n = 10$.

(b) Prove that $\displaystyle\sum_{k=1}^{n}(2k - 1) = n^2$ by splitting the sum.

Exercise 2.4 *(▶ ⋆⋆ core — composition and decomposition)*. Let $f(x) = \sqrt{x}$ and $g(x) = 1 - x^2$.

(a) Find $(f \circ g)(x)$ and its domain, and $(g \circ f)(x)$ and its domain.

(b) Express $h(x) = (2x+1)^5$ and $p(x) = \dfrac{1}{\sqrt{x^2+1}}$ each as a composition of simpler functions.

Exercise 2.5 *(▶ ⋆⋆ core — the same rule, different verdicts)*. Consider the rule $f(x) = x^2$.

(a) As a map $\mathbb{R} \to \mathbb{R}$, show it is neither injective nor surjective.

(b) Show that as a map $[0, \infty) \to [0, \infty)$ it is bijective, and write down its inverse.

Exercise 2.6 *(⋆⋆ core — an inverse via $y = x^3$)*.

(a) Explain why $f(x) = x^3$ is bijective on $\mathbb{R}$, and find f^{-1}.

(b) Find the inverse of $g(x) = (x-1)^3 + 2$ and verify that $g\big(g^{-1}(x)\big) = x$.

Exercise 2.7 *(▶ ⋆⋆ core — graph transformations)*.

(a) Obtain $g(x) = (x-2)^2 + 1$ from $y = x^2$ by a sequence of transformations; state the vertex and sketch both curves.

(b) Obtain $h(x) = 3 - \sqrt{x+2}$ from $y = \sqrt{x}$. State the domain, the transformations used, and whether h is increasing or decreasing.

Exercise 2.8 *(⋆⋆ core — even and odd parts (maths))*.

(a) Prove that the product of two odd functions is even.

(b) Show that any function f on a domain symmetric about 0 can be written $f = E + O$ with $E(x) = \frac{1}{2}\big(f(x) + f(-x)\big)$ even and $O(x) = \frac{1}{2}\big(f(x) - f(-x)\big)$ odd. Apply this to $f(x) = e^x$.

Exercise 2.9 *(⋆⋆ core — reading the piecewise ReLU)*. Recall $\mathrm{ReLU}(x) = \max(0, x)$.

(a) Write ReLU as a piecewise formula, and evaluate it at $x = -3$, 0, 2.

(b) Show that $\mathrm{ReLU}(x) = \frac{1}{2}\big(x + |x|\big)$.

(c) Hence express $|x|$ in terms of $\mathrm{ReLU}(x)$ and $\mathrm{ReLU}(-x)$.

Exercise 2.10 *(⋆⋆ core — exponentials and logarithms)*. Solve for x (exact values).

(a) $2^x = 10$, (b) $3\,e^{2x} = 12$, (c) $\ln 8 - \ln 2 + \ln \frac{1}{2}$ (simplify to a single log).

Exercise 2.11 *(▶ ⋆⋆ core — trigonometric equations)*. Solve on $[0, 2\pi)$, giving exact answers in radians.

(a) $\cos x = \frac{1}{2}$, (b) $\tan x = 1$, (c) $2 \sin x - 1 = 0$.

Exercise 2.12 *(▶ ⋆⋆ core — the sigmoid as a probability (data science))*. The logistic (sigmoid) function is $\sigma(x) = \dfrac{1}{1 + e^{-x}}$.

(a) Compute $\sigma(0)$ and explain why $0 < \sigma(x) < 1$ for every real x.

(b) Verify the symmetry $\sigma(-x) = 1 - \sigma(x)$.

(c) In logistic regression a real-valued score s is turned into a class probability $\sigma(s)$. If $s = \ln 4$, find the probability. What score gives probability $\frac{1}{2}$?

Exercise 2.13 *(▶ ⋆⋆ core — a potential barrier from steps (physics))*. The rectangular potential barrier of height $V_0 > 0$ and width $2a$ is $V(x) = V_0$ for $-a \le x \le a$ and $V(x) = 0$ otherwise. Throughout, H is the Heaviside step, with the convention $H(0) = 1$ (so $H(x) = 1$ for $x \ge 0$ and $H(x) = 0$ for $x < 0$).

(a) What is the range of V? Where is V discontinuous?

(b) Show that $V(x) = V_0\big[H(x+a) - H(x-a)\big]$, and check the value at $x = 0$ and at $x = 2a$.

Exercise 2.14 (▶ $\star\star$ *core — the inverse-trig function* arctan)**.** arctan is the inverse of tan restricted to the branch $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(a) State the maximal domain and the range of arctan.

(b) Explain why the lines $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes, and why arctan is an odd function.

(c) Evaluate $\arctan 0$ and $\arctan 1$, and sketch $y = \arctan x$.

Exercise 2.15 ($\star\star$ *core — sketching the sigmoid* (data science))**.** For the logistic (sigmoid) function $\sigma(x) = \dfrac{1}{1+\mathrm{e}^{-x}}$, produce a careful hand sketch. On your graph mark: the value $\sigma(0)$, the two horizontal asymptotes, the direction of monotonicity, and the centre of symmetry. Justify each feature in one line.

3 Further practice (home)

A larger bank for independent study, rising from routine to harder, with conceptual checks (true/false and spot-the-error) before a closing block of $\star\star\star$ challenge challenge / extension problems.

Exercise 3.1 ($\star\star$ *core — notation round-up*)**.**

(a) Write $\{\, x \in \mathbb{R} \;:\; 1 < 2x - 3 \le 7 \,\}$ as an interval.

(b) Write $[-2, 0) \cup (0, 2]$ in set-builder notation in one line.

(c) Evaluate $\displaystyle\sum_{k=0}^{4} 2^k$ and $\displaystyle\prod_{k=1}^{n} \frac{k+1}{k}$.

Exercise 3.2 ($\star\star$ *core — a rational inequality*)**.** Solve $\dfrac{x}{x+1} < 2$.

Exercise 3.3 ($\star\star$ *core — maximal domains, harder*)**.** State the maximal domain of each.

$$\text{(a) } f(x) = \ln(x^2 - 4), \qquad \text{(b) } g(x) = \frac{\sqrt{x^2-1}}{x-3}, \qquad \text{(c) } h(x) = \arcsin(x-1).$$

Exercise 3.4 ($\star\star$ *core — compositions with reciprocals*)**.** Let $f(x) = \dfrac{1}{x}$ and $g(x) = x + 1$.

(a) Find $f \circ g$, $g \circ f$ and $f \circ f$, each with its domain.

(b) Decompose $F(x) = \sqrt{\ln x}$ into simpler functions and state its domain.

Exercise 3.5 ($\star\star$ *core — inverses, including a Möbius map*)**.** Find the inverse (with domain) of

$$\text{(a) } f(x) = 2 + \sqrt{x-1}, \qquad \text{(b) } g(x) = \frac{3x+1}{x-2}.$$

Exercise 3.6 ($\star\star$ *core — classify the maps*)**.** Decide injective / surjective / bijective.

(a) $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 - x$.

(b) $f\colon (-\infty, 0] \to [0, \infty)$, $f(x) = x^2$.

Exercise 3.7 *(⋆⋆ core — symmetry zoo)*. Classify each as even, odd or neither; note any obvious period.

$$x^5 \sin x, \qquad x\,|x|\,, \qquad \mathrm{e}^{-x^2}, \qquad x + \cos x, \qquad |\sin x|\,.$$

Exercise 3.8 *(⋆⋆ core — transforming standard graphs)*.

(a) From $y = \dfrac{1}{x}$, describe the graph of $y = \dfrac{1}{x-2} + 1$, giving both asymptotes.

(b) From $y = \sin x$, describe the graph of $y = 2\sin(x) - 1$, giving its range.

Exercise 3.9 *(⋆⋆ core — log and exponential equations)*. Solve for x.

$$\text{(a)}\ \log_2 x + \log_2(x-2) = 3, \qquad \text{(b)}\ \mathrm{e}^{2x} - 5\mathrm{e}^{x} + 6 = 0.$$

Exercise 3.10 *(⋆⋆ core — more trigonometric equations)*. Solve on $[0, 2\pi)$.

$$\text{(a)}\ 2\cos^2 x - 1 = 0, \qquad \text{(b)}\ \sin x = \cos x.$$

Exercise 3.11 *(⋆⋆ core — the logit* (data science)*)*. For the sigmoid $\sigma(x) = \dfrac{1}{1+\mathrm{e}^{-x}}$ (range $(0,1)$):

(a) Show that its inverse is the *logit* $\sigma^{-1}(p) = \ln \dfrac{p}{1-p}$, $0 < p < 1$.

(b) A classifier outputs probability $p = 0.8$. What score s produced it?

Exercise 3.12 *(⋆⋆ core — a pulse from steps* (physics)*)*. Using the Heaviside step H, write a single formula for the signal that is 0 for $t < 0$, equal to 5 for $0 \le t < 2$, and 0 for $t \ge 2$. Check it at $t = 1$ and $t = 3$.

Exercise 3.13 *(⋆⋆ core — hyperbolic identities)*. Work directly from $\sinh x = \frac{1}{2}(\mathrm{e}^{x} - \mathrm{e}^{-x})$ and $\cosh x = \frac{1}{2}(\mathrm{e}^{x} + \mathrm{e}^{-x})$.

(a) Prove $\cosh^2 x - \sinh^2 x = 1$.

(b) Prove $\sinh(2x) = 2\sinh x \cosh x$.

Exercise 3.14 *(⋆⋆ core — finding ranges)*. Find the range of each function.

$$\text{(a)}\ f(x) = \frac{x^2}{x^2+1}, \qquad \text{(b)}\ g(x) = 2\sin x + 1.$$

Exercise 3.15 *(⋆⋆ core — true or false?)*. State whether each is true or false, with a one-line reason.

(a) $\infty \in \mathbb{R}$.

(b) Every injective function $\mathbb{R} \to \mathbb{R}$ is surjective.

(c) $f \circ g = g \circ f$ for all functions f, g.

(d) The maximal domain of $\ln(x^2)$ is all of $\mathbb{R}$.

(e) Every periodic function is bounded.

Exercise 3.16 *(⋆⋆ core — spot the error)*. Each "solution" below contains a mistake. Find and correct it.

(a) "$-2x > 6$, so $x > -3$."

(b) "The maximal domain of $\sqrt{x-2}$ is $x \le 2$."

(c) "$|x| = -2$ has the solutions $x = \pm 2$."

Exercise 3.17 *($\star\star\star$ challenge — irrationality of $\sqrt{3}$)*. Prove that $\sqrt{3}$ is irrational, imitating the proof for $\sqrt{2}$ in the notes.

Exercise 3.18 *($\star\star\star$ challenge — the Gauss sum)*. Prove $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ by writing the sum forwards and backwards and adding, with no appeal to induction.

Exercise 3.19 *($\star\star\star$ challenge — monotonic $\Rightarrow$ injective, but not conversely)*.

(a) Prove that a strictly increasing function is injective.

(b) Give an injective function on $[0, 2]$ that is *not* monotonic.

Exercise 3.20 *($\star\star\star$ challenge — sigmoid and* tanh*)*. Show that $\sigma(x) = \dfrac{1 + \tanh(x/2)}{2}$.